R13

Code No: 118BJ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, May - 2019
EMBEDDED SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a)	Discuss the use of interrupts in 8051 microcontroller.	[2]
b)	What are the requirements of embedded system?	[3]
c)	Explain the need of Decimal Arithmetic instruction in 8051.	[2]
d)	Discuss about Assembly Language Programming Tools.	[3]
e)	Explain the necessity of RTOS in embedded system?	[2]
f)	Explain about the function of Message Queues in RTOS.	[3]
g)	Define hard-real time and soft-real time embedded systems.	[2]
h)	Write a short note on Linker in embedded software	[3]
i)	List the features of ARM7.	[2]
j)	Explain different data processing instructions in detail.	[3]

PART - B

(50 Marks)

2. Explain in detail about embedded system design process with examples. [10]

OR

3.a) What are the different types of memories used in 8051 Micro controller?
b) Explain the interrupt handling mechanism of 8051. [5+5]

4. Explain the various Arithmetic instructions of 8051, with examples. [10]

OR

5.a) Describe
i) Interrupt priority
ii) Interrupt deadline
b) Explain about data transfer and logical instructions with examples. [4+6]

6.a) Explain in detail A/D convertor interfacing with a microcontroller.
b) Explain the concept of semaphore. How these help us in solving the shared-data problem in embedded system? [6+4]

OR

7.a) Explain about the standard 8-bit UART mode of serial communication.
b) What is a Task and briefly explain about the task scheduling mechanism? [4+6]

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8. Describe the main features of Micro Controller open source RTOS. [10]

OR

9. Write about Embedded software Development tools briefly. [10]

10. Explain the necessity of Architecture Revisions in ARM processor. [10]

OR

11. Describe the basic concepts CAN bus protocol. [10]

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